

MMAE 545 – Project 2

Minimum-Weight Aluminum Bladed Disk Design

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Introduction

The objective of this project is to design a minimum-weight aluminum bladed disk assembly operating at 4,500 RPM. Two structural constraints govern the design: (1) a minimum factor of safety of 2.0 against yielding, giving a maximum allowable von Mises stress of 17,500 psi ($\sigma_y = 35,000$ psi), and (2) a maximum radial blade tip deflection of 0.015 in. The disk has 24 blades; blade tips reach $r = 24.0$ in, the outer hub radius is 15.0 in, and the shaft spans 8.0 in with fixed ends. Blade width $3.00 \leq w \leq 4.25$ in and thickness $0.05 \leq t \leq 0.75$ in may taper linearly with largest values at the root. The hub region $r = 2.0$ to 14.75 in is the design freedom zone (max axial width 5.0 in, fully solid). Linear static FEA was performed in ANSYS Workbench 2025 R2 (IPS units), followed by prestressed modal analysis at 4,500 and 9,000 RPM.

Modeling and Optimization Procedure

Pre-Analysis Estimate

Centrifugal blade root stress was estimated before model construction:

$$\sigma \approx \frac{\rho \omega^2 (r_{\text{tip}}^2 - r_{\text{root}}^2)}{2} \quad (1)$$

At $\omega = 471.24$ rad/s, $\rho = 2.457 \times 10^{-4}$ lb·s²/in⁴: blade inertia alone contributes $\approx 9,590$ psi – 55% of the allowable – before hub load or fillet concentration. Stress governs; an outside-in strategy was adopted to minimize blade mass first.

Geometry Construction

A single connected solid sector was built in ANSYS DesignModeler. Gravitational effects were neglected as negligible relative to centrifugal body forces at 4,500 RPM.

Shaft Hollow cylinder, $r_i = 1.0$ in, $r_o = 2.0$ in, $L = 8.0$ in. Non-design region. A fillet at the shaft-to-hub junction ($r = 2.0$ in) suppresses stress concentration at that interface.

Hub ($r = 2.0$ to 14.75 in). Sole legal zone for material removal. Final hub axial thickness: 0.0484 in; hub mount fillet radius: 0.159 in. A radial chamfer provides a continuous load-path transition from blade root to shaft. Maximum axial width was confirmed below 5.0 in at all radial locations including the chamfer transition. A shallow blind pocket was introduced in the confirmed low-stress mid-hub region to reduce mass.

Blade Mounting Platform ($r = 14.75$ to 15.0 in). Fixed 5.0 in axial width through the full 0.25 in radial depth. Non-design region.

Blade ($r = 15.0$ to 24.0 in). Lofted between root and tip rectangular profiles. Thickness and width set to minimum legal values ($t = 0.05$ in, $w = 3.0$ in) to minimize centrifugal load. A compound

chamfer-and-fillet at the blade base controls tip deflection via gusset action; the fillet (0.185 in radius) eliminates the root stress singularity.

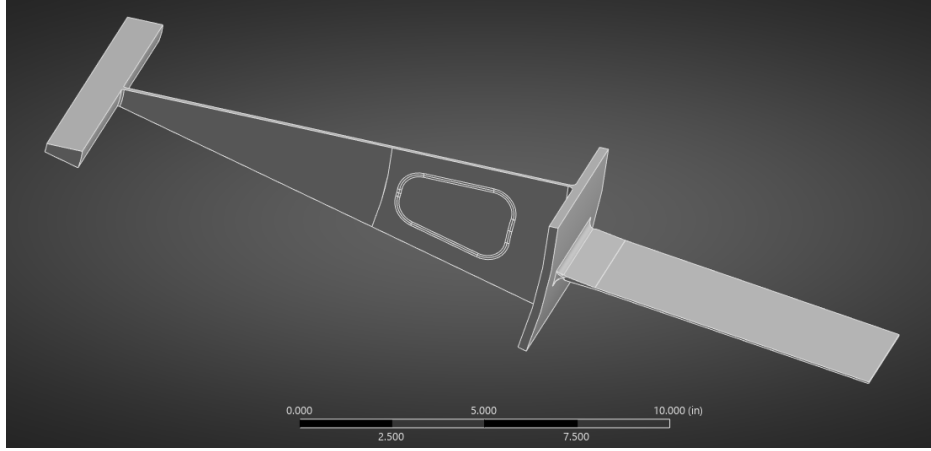


Figure 1: Final sector geometry in ANSYS DesignModeler (1/24 of full assembly).

Finite Element Model

Material properties: $E = 10.5 \times 10^6$ psi, $\nu = 0.34$, $\rho = 0.095$ lb/in³, $\sigma_y = 35,000$ psi. SOLID187 (10-node quadratic tetrahedral) elements were used. A linear static analysis was performed with large-deflection effects disabled.

The 15° sector represents 1/24 of the assembly. Both geometry and centrifugal loading are circumferentially periodic, satisfying cyclic symmetry requirements. Cyclic symmetry BCs were applied to the two radial sector faces. Fixed supports applied to both shaft end faces. Cyclic symmetry boundary conditions were applied to the sector faces. Rotational velocity of 4,500 RPM was applied about the axis of rotation.

Element quality metrics confirmed acceptable skewness and aspect ratios, where face meshing as applied. Radial tip deflection was scoped to the blade tip (face at $r = 24.0$ in) using cyclical X -direction, ensuring true radial extraction at all circumferential positions.

Mesh Convergence

Mesh independence was verified across five blade face-sizing levels (Table 1). Convergence between the two finest levels:

$$\Delta\sigma = \frac{|15,800 - 16,050|}{15,800} \times 100 = 1.58\% \quad \Delta\delta = \frac{|0.014984 - 0.014983|}{0.014984} \times 100 = 0.007\% \quad (2)$$

Both are below the 2% criterion. Non-monotonic stress variation at coarser levels is attributed to resolution sensitivity at the blade root fillet, where coarse meshes under-resolve peak gradients.

Table 1: Mesh convergence study. Criterion: $\leq 2\%$ change between successive levels.

Level	Blade Face Size (in)	Mass (lbm)	$\sigma_{\max}$ (psi)	δ_{tip} (in)
1(fine)	0.0475	1.1781	15,800	0.014984
2	0.0925	1.1781	16,050	0.014983
3	0.1375	1.1781	16,518	0.014984
4	0.1825	1.1781	16,871	0.014983
5(coarse)	0.2275	1.1781	16,778	0.014983

Optimization Procedure

Topology optimization identified low-stress hub regions and confirmed where material removal was structurally justified. Parametric optimization was then conducted in ANSYS DesignXplorer using the Adaptive Multiple-Objective (AMO) algorithm: minimize sector mass; constrain von Mises stress $\leq 17,300$ psi (200 psi buffer); constrain radial tip deflection ≤ 0.0145 in. Upper bounds without targets allow the optimizer to approach each limit naturally. Following parametric optimization, a blind pocket was introduced in the confirmed low-stress mid-hub region; a re-solve confirmed maximum stress increased by less than 2%, validating the removal.

Results and Discussion

Final Design Summary

The final sector mass is 1.1781 lbm, representing $\frac{1}{24}$ of the full assembly including the proportional shaft contribution. Total assembly mass: $1.1781 \times 24 = 28.27$ lbs. Constraint utilization:

$$\text{Stress utilization} = \frac{16,511}{17,500} = 94.3\% \quad \text{Deflection utilization} = \frac{0.014983}{0.015} = 99.9\%$$

All design constraints are satisfied (Table 2).

Table 2: Final design results at 4,500 RPM.

Parameter	Value	Limit	Status
Total assembly weight (lbs)	28.274	Minimize	–
Sector mass (lbm)	1.1781	–	–
Max radial tip deflection (in)	0.014983	≤ 0.015	PASS
Max von Mises stress (psi)	16,511	$\leq 17,500$	PASS
Factor of safety ($\sigma_y/\sigma_{\max}$)	2.12	≥ 2.0	PASS

Stress Distribution

The von Mises stress distribution at 4,500 RPM is shown in Figure 2. The maximum stress of 16,511 psi occurs at the blade root lateral face where centrifugal loading produces peak bending stress amplified by the geometric transition. $FoS = 35,000/16,511 = 2.12$, satisfying the minimum of 2.0. Hub stresses are substantially lower, confirming material removal occurred only in non-load-bearing regions. All re-entrant edges are filleted; no stress singularities are present.

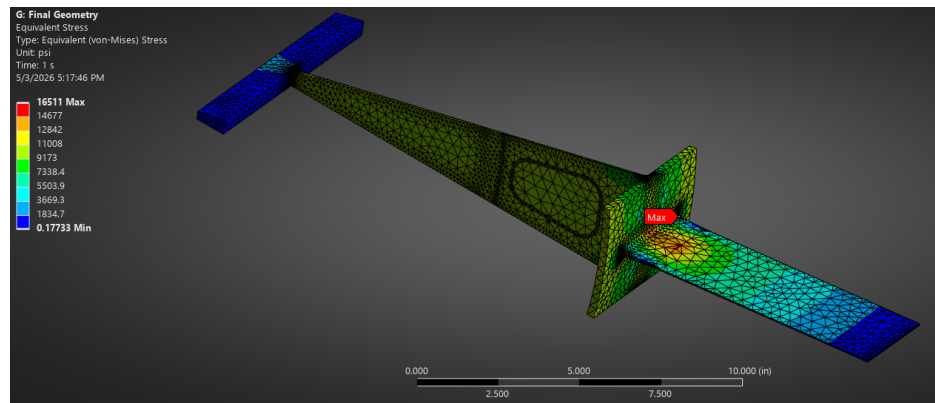


Figure 2: Von Mises equivalent stress at 4,500 RPM. Maximum = 16,511 psi at blade root lateral face. Factor of safety = 2.12.

Radial Deflection

The deflection constraint applies strictly at the blade tip ($r = 24.0$ in); deformation elsewhere does not constitute a violation. Maximum radial tip deflection is 0.014983 in (99.9% of the 0.015 in limit); mesh convergence across all levels confirms this value is numerically stable. The global maximum deformation is 0.015725 in at the blade mounting platform outer edge ($r = 15.0$ in) due to local geometric stiffness variation; this location is not constrained. Deflection generally increases along the blade span toward the tip, consistent with centrifugal loading physics.

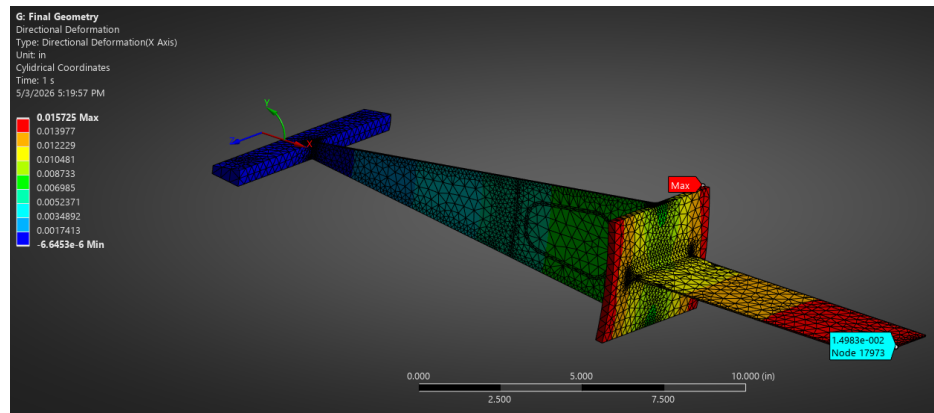


Figure 3: Radial tip deflection at 4,500 RPM. Maximum at blade tip ($r = 24.0$ in) = 0.014983 in (99.9% of 0.015 in limit). Constraint is satisfied.

Natural Frequencies at 4,500 RPM

The first four undamped natural frequencies from the prestressed modal analysis at 4,500 RPM are listed in Table 3. A harmonic index sweep from 0 to 12 was performed ($N/2 = 12$ for a 24-blade assembly); mode shapes confirmed axisymmetric ($h = 0$) and nodal diameter modes.

Table 3: First four undamped natural frequencies at 4,500 RPM (prestressed modal).

Mode	h	Freq. (Hz)	Description
1	0	51.125	1st Flexural Bending – axisymmetric out-of-plane wag
2	1	76.546	1st Nodal Diameter Bending – side-to-side with twist
3	0	122.27	2nd Flexural Bending – axisymmetric up-and-down
4	2	124.53	2nd Nodal Diameter Bending – side-to-side with twist

Results at 9,000 RPM

Results at 9,000 RPM are in Table 4. No constraints apply at this speed. Stress at 9,000 RPM is 66,045 psi = $4.00\times$ the 4,500 RPM value, confirming $\sigma \propto \omega^2$ scaling. This exceeds yield (35,000 psi); sustained operation is not intended. Natural frequencies increase due to stress stiffening (geometric stiffening from centrifugal pre-stress); this increase is nonlinear and significantly less than ω^2 scaling.

Table 4: Comparison of results at 4,500 RPM and 9,000 RPM.

Parameter	4,500 RPM	9,000 RPM
Max radial tip deflection (in)	0.014983	0.059932
Max von Mises stress (psi)	16,511	66,045*
Natural frequency – Mode 1 (Hz)	51.125	95.36
Natural frequency – Mode 2 (Hz)	76.546	151.52
Natural frequency – Mode 3 (Hz)	122.27	169.37
Natural frequency – Mode 4 (Hz)	124.53	231.47

*Exceeds yield strength (35,000 psi); sustained operation not intended.

Discussion

The design operates near both constraint limits, with 99.9% deflection utilization and 94.3% stress utilization. The deflection constraint is governing; any further mass reduction would violate the 0.015 in tip deflection limit before reaching the stress limit. This indicates a constraint-active optimum. The outside-in strategy effectively reduced centrifugal loading by minimizing blade thickness, allowing the hub to be thinned without exceeding stress limits. Local geometric features (fillet and chamfer) controlled stress concentration and stiffness, while the blind hub pocket removed non-load-bearing material with negligible impact ($< 2\%$) on peak stress.

Stress stiffening is evident: natural frequencies increase with rotational speed due to centrifugal pre-stress. At 4,500 RPM, the fundamental frequency (51.125 Hz) remains separated from the primary excitation frequency (75 Hz), and no excitation harmonics coincide with the first four modes.

Conclusion

A minimum-weight aluminum bladed disk was designed using cyclic symmetry FEA and parametric optimization. The final assembly mass is 28.27 lbs. At 4,500 RPM, the design satisfies all constraints: maximum stress is 16,511 psi ($FoS = 2.12$) and radial tip deflection is 0.014983 in. Mesh convergence confirms solution independence. The design achieves a constraint-active optimum, with deflection governing. Further weight reduction is not feasible without violating structural limits. Prestressed modal analysis shows increasing natural frequencies with rotational speed due to stress stiffening, with no resonance observed in the first four modes.

Appendix

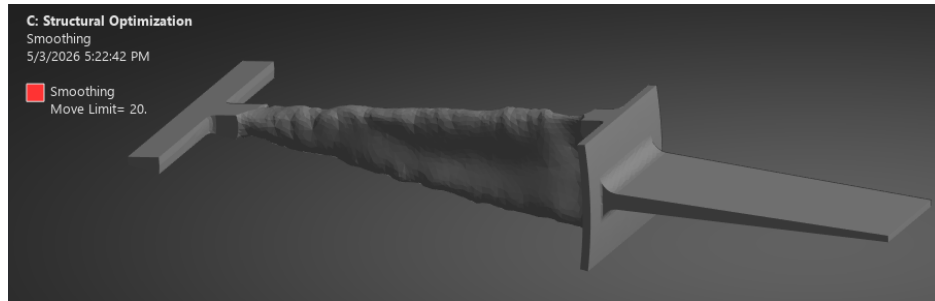


Figure A1: Topology optimization result. Retained material distribution confirms primary load paths and guides blind pocket placement in hub.

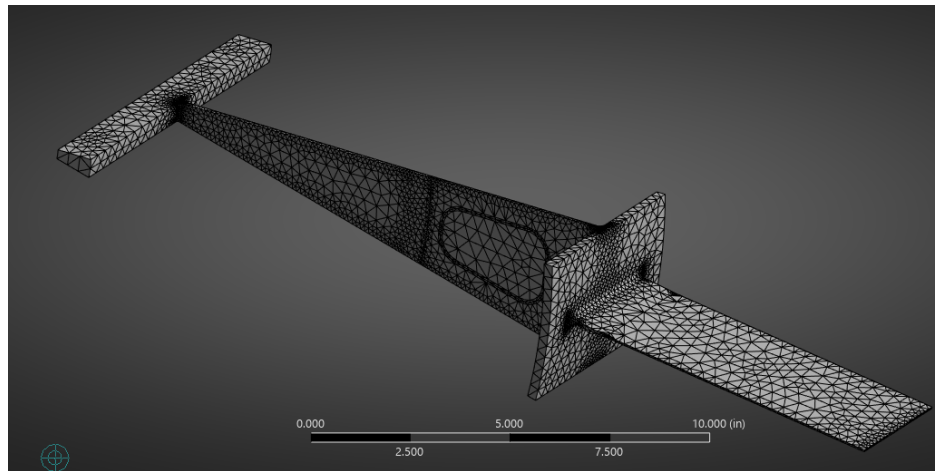


Figure A2: Final FE mesh.

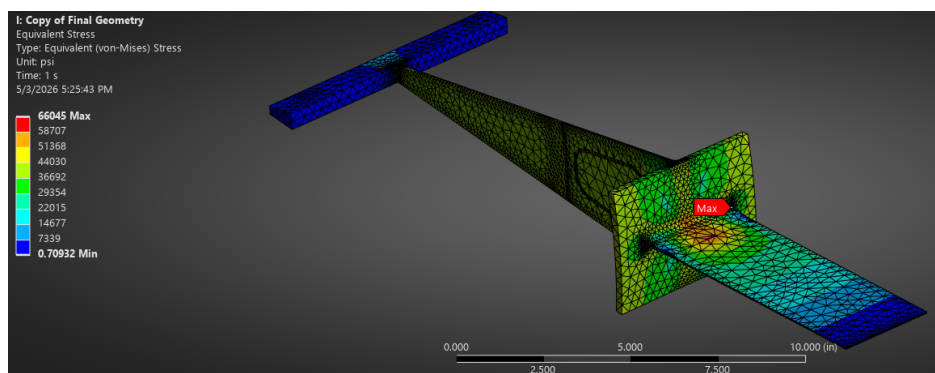


Figure A3: Von Mises stress at 9,000 RPM (reference only). Maximum = 66,045 psi, exceeding yield strength (35,000 psi); sustained operation at this speed is not intended.